

Detecting Differential Item Functioning Using Posterior Predictive Model Checking: A Comparison of Discrepancy Statistics

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This study proposes a new Bayesian differential item functioning (DIF) detection method using posterior predictive model checking (PPMC). Item fit measures including infit, outfit, observed score distribution (OSD), and Q1 were considered as discrepancy statistics for the PPMC DIF methods. The performance of the PPMC DIF method was evaluated via a Monte Carlo simulation manipulating sample size, DIF size, DIF type, DIF percentage, and subpopulation trait distribution. Parametric DIF methods, such as Lord's chi-square and Raju's area approaches, were also included in the simulation design in order to compare the performance of the proposed PPMC DIF methods to those previously existing. Based on Type I error and power analysis, we found that PPMC DIF methods showed better-controlled Type I error rates than the existing methods and comparable power to detect uniform DIF. The implications and recommendations for applied researchers are discussed.

In educational and psychological assessments, the detection of differential item functioning (DIF) is one of the most important procedures in research and operational settings. DIF is defined as an expected item score difference across multiple groups after the scales and scores are calibrated and linked on the same metric (Holland & Wainer, 1993; Lord, 1980). Several pieces of past literature have exclusively discussed factors that cause DIF in the contexts of academic performance (Camilli, 1992; Hambleton & Murphy, 1992; Penfield & Lam, 2000; Zwick et al., 1993), national and international surveys (Rutkowski & Svetina, 2014; Zwick & Ercikan, 1989), attitude (Dodeen & Johanson, 2003; Meyer et al., 2004), personality (Stark et al., 2004; Zickar & Robie, 1999), and health assessments (Andrich & Hagquist, 2015; Teresi & Fleishman, 2007). The presence of DIF items not only causes psychometric problems, such as biasing group ability estimates or distorting the reliability and validity of scales (Ackerman, 1992; Millsap & Everson, 1993; Zwick et al., 1997), it also creates fairness issues in testing and adverse impacts in high-stakes settings (Penfield & Camilli, 2007; Santelices & Wilson, 2010). Thus, identifying DIF items with reliable methods is critical in the development and validation of educational and psychological scales.

In psychometric literature, several DIF methods have been developed and discussed. For example, Lord's chi-square (Lord, 1980), Raju's differential functioning of items and tests (DFIT; Raju et al., 1995), and the likelihood ratio test (LRT; Thissen et al., 1993) were previously developed as parametric DIF methods, and their

effectiveness was systematically examined via simulation studies (e.g., Cohen et al., 1996; Oshima et al., 2006). Nonparametric DIF methods such as logistic regression (Swaminathan & Rogers, 1990), Mantel-Haenszel (MH; Mantel & Haenszel, 1959), and the simultaneous item bias test (SIBTEST; Shealy & Stout, 1993) were also developed and widely used in operational and research settings. Finally, under the structural equation modeling (SEM) framework, multiple-group confirmatory factor analysis (MGCFA) and multiple-indicator multiple-cause (MIMIC) methods have been incorporated to detect uniform and nonuniform DIF (Kim & Yoon, 2011; Woods, 2009a; Woods & Grimm, 2011).

Although several DIF methods have been discussed previously, a majority of them have been mainly developed under the classical statistic framework. Recently, Bayesian statistics have received increasing attention in the item response theory (IRT) literature because (1) it takes prior information into account for increasing the accuracy and reliability of measurement, (2) it does not require complex analytic forms of the probability functions to estimate model parameters, (3) it is flexible enough to summarize the posterior distribution of model parameters by computing mean (i.e., expected a posteriori [EAP]), mode (i.e., maximum a posteriori [MAP]) or median, and (4) Bayesian estimation such as Markov chain Monte Carlo (MCMC) generally outperforms conventional estimation such as marginal maximum likelihood (MML) when the sample size is relatively small.

With these advantages, Bayesian statistics have been intensively applied for psychometric analyses such as model estimation (Béguin & Glas, 2001; Fox & Glas, 2001; Patz & Junker, 1999), model data fit (Sinharay, 2005; Sinharay et al., 2006), dimensionality (Levy, 2011; Levy et al., 2009), and measurement invariance (Verhagen & Fox, 2013; Van de Schoot et al., 2015). Despite the adaptations of several psychometric applications under the Bayesian framework, only a handful of psychometric research has explored the Bayesian DIF method previously. For example, Zwick et al. (1999, 2000) proposed the empirical Bayes (EB) approach to enhance the performance of MH statistics for detecting DIF. They used previous data to construct the prior distribution and made statistical inferences using the posterior distribution of DIF parameters. They concluded that the MH statistic with the EB approach was effective when the sample size was relatively small. Later, Sinharay et al. (2009) extended the MH statistic by using the full Bayesian (FB) approach to construct the prior distribution and showed the outperformance via a simulation study. Furthermore, Wang et al. (2008) developed the MCMC algorithm for the MH statistic using the testlet response theory model and showed the advantages of MCMC for DIF detection. Most recently, Joo et al. (2022) showed Bayesian DIF detection methods using Bayes factor (BF) and deviance information criterion (DIC) for an ideal point IRT model.

Although previous research considerably advanced Bayesian DIF methods and provided important insights for applied researchers and practitioners, the studies were still limited to nonparametric DIF methods, and a Bayesian parametric DIF method has not yet been developed. More importantly, Sinharay et al. (2006) previously illustrated the use of posterior predictive model checking (PPMC) for assessing IRT model fit and showed the flexibility of PPMC by implementing discrepancy statistics. PPMC is useful for constructing an empirical distribution of any

statistics (i.e., discrepancy statistics) that do not follow an asymptotic distribution. Based on the empirical distribution, one can make Bayesian statistical inferences similar to classical statistics by computing the posterior predictive p-value (PPP). For example, Levy et al. (2009) adapted the PPMC procedure to investigate empirical distributions of several item-pair statistics, including odds ratio (OR), MH, and Q3 statistics, and illustrated the use of PPMC for scale dimensionality assessment. We believe that the PPMC approach can be extended for detecting DIF by incorporating Bayesian multigroup estimation along with appropriate item-level fit statistics. To the best of the authors' knowledge, no research has proposed such a method or explored the performance of different discrepancy statistics in the context of DIF detection. Therefore, the purpose of the current study is to propose a new DIF method using the PPMC approach and evaluate the performance of various item-level fit measures as discrepancy statistics. The remainder of the paper describes the PPMC procedure for detecting DIF, the simulation study design, and the results, implications and limitations of the study.

DIF Analysis Using the PPMC Procedure

Under the IRT framework, the two-parameter logistic model (2PL) has been widely used to model dichotomous item response data in applied settings. The 2PL model is defined as:

$$P(X_{ij} = 1 | \theta_i, \alpha_j, \beta_j) = \frac{\exp(\alpha_j(\theta_i - \beta_j))}{1 + \exp(\alpha_j(\theta_i - \beta_j))}, \quad (1)$$

where X_{ij} is an item response for a j th item ($j = 1, \dots, J$) from an i th examinee ($i = 1, \dots, N$), α_j is a j th item discrimination parameter, β_j is a j th item difficulty parameter, and θ_i is an i th examinee's ability parameter. For the multigroup case, the conditional probability function can be formulated as:

$$P(X_{ijg} = 1 | \theta_{ig}, \alpha_{jg}, \beta_{jg}) = \frac{\exp(\alpha_{jg}(\theta_{ig} - \beta_{jg}))}{1 + \exp(\alpha_{jg}(\theta_{ig} - \beta_{jg}))} \text{ and } \theta_{ig} \sim N(\mu_{\theta_g}, \sigma_{\theta_g}^2), \quad (2)$$

where X_{ijg} is an item response for a j th item from an i th examinee for a group g ($g = 1, \dots, G$), and parameters of the model have the subscript g for a specific group. Note that μ_{θ_g} and $\sigma_{\theta_g}^2$ are the group-specific mean and variance parameters for the group population. The model parameters described in Equation 2 can be estimated using MCMC and the detailed procedure is described in the Appendix 1.

PPMC (Gelman et al., 1996) was developed as a Bayesian method for model fit assessment by constructing the posterior predictive distribution. The posterior predictive distribution refers to the predictive distribution of replicated data conditional on the observed data (Sinharay et al., 2006). To describe the PPMC procedure, suppose ω is a set of parameters of the IRT model and X is observed data. Based on the fundamental principle of Bayes' Theorem, the posterior distribution of ω , given the likelihood function $P(X|\omega)$ and the prior distribution $P(\omega)$, can be formulated as: $P(\omega|X) \propto P(X|\omega)P(\omega)$. Once the posterior distribution is obtained, the posterior predictive distribution of the replicated data X^{rep} can then be obtained from:

$$P(X^{rep}|X) = \int P(X^{rep}|\omega) P(\omega|X) d\omega, \quad (3)$$

where $P(\omega|X)$ is the posterior distribution of ω and $P(X^{rep}|\omega)$ is the conditional probability of the replicated data X^{rep} , given parameters ω . Note that the replicated data X^{rep} is considered “predicted” or “future” observed data, and that by examining the discrepancy between the observed and replicated data, one can “check” the model data fit (Rubin, 1984). The observed and predicted data can be summarized with a discrepancy statistic D (a.k.a. test statistic), and the difference between the posterior distribution of $D(X, \omega)$ and the posterior predictive distribution of $D(X^{rep}, \omega)$ indicates model misfit (Sinharay et al., 2006). Although the PPMC method was originally developed to evaluate Bayesian model fit with diagnostic plots (e.g., QQ or residual plots), quantitative measures can also be obtained from the posterior predictive p - (PPP) value. The PPP value is computed as:

$$P(D(X^{rep}, \omega) \geq D(X, \omega) | X) = \int_{D(X^{rep}, \omega) \geq D(X, \omega)} P(X^{rep} | \omega) P(\omega | X) dX^{rep} d\omega. \quad (4)$$

In the process of MCMC estimation, Equations 3 and 4 can be approximated simultaneously as follows:

1. Markov samples for model parameters ω^{rep} from the MCMC estimation are first used to simulate the replicated data X^{rep} .
2. The observed and replicated data are then summarized with the discrepancy statistics, with $[D(X, \omega^{rep})$ and $D(X^{rep}, \omega^{rep})]$ or without $[D(X)$ and $D(X^{rep})]$ the parameter samples ω^{rep} depending on the discrepancy statistics.
3. Finally, the PPP can be obtained by computing the proportion of $D(X^{rep}, \omega^{rep})$ that exceeds or is equal to $D(X, \omega^{rep})$.

A PPP value closer to .5 indicates that the observed data fits the model, and a value closer to 0 or 1 indicates that the model does not fit the observed data. Although it is practically possible to make a statistical inference with the PPP value by applying $\alpha/2$ and $1 - \alpha/2$ as the tail-area probabilities, there is an important difference between PPP and classical p value. PPP does not always follow a uniform distribution when the model correctly fits the data, and the correct model tends to produce a PPP value of .5 more often than expected (Bayarri & Berger, 2000; Sinharay et al., 2006). This indicates that the distribution of PPP varies depending on types of misfit and discrepancy statistics (Berkhof et al., 2000; Levy et al., 2009). Thus, the distributions of PPP should be examined prior to the PPMC hypothesis testing.

In the context of PPMC DIF detection, item parameters are first concurrently calibrated using Bayesian multigroup estimation with equal constraints across groups (both discrimination and difficulty item parameters). The item parameter estimates per group are then individually evaluated using PPMC to detect DIF. More specifically, the PPP values are computed for each item-by-group with any choice of discrepancy statistics, and if the PPP value for a DIF studied item is less than or equal to the predetermined tail-area probability $\alpha/2$ (or greater than or equal to $1 - \alpha/2$), then the studied item is flagged as DIF.

Discrepancy Statistics

For the purpose of PPMC DIF detection, several item-level fit statistics can be incorporated as discrepancy statistics. In this study, we considered infit, outfit, observed score distribution (OSD), and Q1 (Yen, 1981) as discrepancy statistics. We chose item fit as discrepancy statistics in the PPMC framework because we used concurrent calibration to scale multiple group data. In multiple group analysis, measurement invariance or DIF is generally evaluated by imposing an equality constraint on items. If the equal constraints do not fit an item, one can identify the item as DIF. This approach is similar to the Wald or LRT DIF tests that view DIF as an item misfit in the multigroup context (Thissen et al., 1993). In this vein, we considered DIF as a part of item misfit and we adapted the item fit indices as discrepancy statistics under the PPMC framework.

Infit and outfit statistics (Wright & Panchapakesan, 1969) were developed as item fit statistics for IRT models (more specifically, the Rasch model [Rasch, 1966]) based on standardized residuals. The infit and outfit statistics for an item j are computed as:

$$\text{infit}_j = \frac{\sum_{i=1}^N SD_{ij}^2 \left[\frac{X_{ij} - P(X_{ij})}{SD_{ij}} \right]^2}{\sum_{i=1}^N SD_{ij}^2}, \quad (5)$$

$$\text{outfit}_j = \frac{1}{N} \sum_{i=1}^N \left[\frac{X_{ij} - P(X_{ij})}{SD_{ij}} \right]^2, \quad (6)$$

where SD is computed as:

$$SD_{ij} = \sqrt{[0 - P(X_{ij})]^2 [1 - P(X_{ij})] + [1 - P(X_{ij})]^2 P(X_{ij})}. \quad (7)$$

Infit and outfit are known to be sensitive to large residuals. In other words, a small number of cases where high ability examinees answer easy items incorrectly, or low ability examinees respond correctly to difficult items, will result in large infit and outfit values (Smith, 1991). Infit and outfit are commonly reported in not only Rasch model software such as WINSTEPS (Linacre, 2010) but also extended model software such as a *mirt* package in R (Chalmers, 2012). Note that infit and outfit statistics were originally developed based on the Rasch model specifically but the statistics are now commonly applied for the non-Rasch IRT models to evaluate item fit in applied settings (e.g., DeMars, 2004; Köhler et al., 2020).

Yen (1981) also developed an item fit statistic for IRT models, referred to as Q1. The Q1 statistic is computed based on 10 equal size examinee strata sorted by the ability estimate ($\hat{\theta}_i$). The Q1 statistic for an item j is computed as:

$$Q_{1j} = \sum_{k=1}^{10} \frac{N_k (O_{jk} - E_{jk})^2}{E_{jk} (1 - E_{jk})}, \quad (8)$$

where N_k is the number of examinees in a stratum k , O_{jk} is the number of examinees getting an item j correct in a stratum k , and E_{jk} is the expected number of examinees getting an item j correct in a stratum k . Yen (1981) showed that the Q1 statistic

asymptotically follows the chi-square distribution with $10 - m$ degrees of freedom, where m is the number of item parameters. However, we carefully note that the 10 strata may not equally work. Particularly, strata at the ends of IRT scores generally include small sample sizes, which results in different degrees of freedom for the chi-square distribution.

In addition, previous PPMC studies (e.g., Sinharay et al., 2006) have considered OSD as a discrepancy statistic. The item-level OSD statistic can be computed as:

$$OSD_j = \frac{[NC_j - E(NC_j)]^2}{E(NC_j)}, \quad (9)$$

where NC_j is the number of examinees getting an item j correct and $E(NC_j)$ is the expected number of examinees getting an item j correct. It has been discussed that the OSD statistic does not follow the chi-square distribution because the expected frequency cannot be computed with the true item parameters, but only estimated from the sample (Ferrando & Lorenzo-Seva, 2001). The statistic, however, is still useful to describe the sample data with the fitted IRT model because the item misfit index generally becomes apparent as the difference between observed and expected frequencies become larger, and more importantly, the reference distribution can be empirically constructed with the posterior predictive samples using the PPMC method (Sinharay et al., 2006).

Method

Simulation Design

A Monte Carlo simulation was conducted to investigate the performance of the PPMC DIF method with different discrepancy statistics (infit, outfit, OSD, and Q1) under various conditions. The simulation design and the corresponding values were chosen based on the previous DIF studies (e.g., Bolt 2002; Cohen et al., 1996; Kim & Cohen, 1992, 1998; Stark et al., 2006; Woods & Grimm, 2011). Five independent simulation factors were manipulated: DIF types (uniform and nonuniform), sample sizes per group (200, 400, 600, 800, and 1000), DIF sizes (defined as item parameter differences; .2, .6, and 1.0 for uniform DIF, .2, .4, and .6 for nonuniform DIF), DIF percentages (10% and 20%), and subpopulation trait distributions (equivalent groups and nonequivalent groups). There were a total of $120 (= 2 \text{ [DIF types]} \times 5 \text{ [sample sizes]} \times 3 \text{ [DIF sizes]} \times 2 \text{ [DIF percentages]} \times 2 \text{ [subpopulation trait distributions]})$ conditions, and for each condition, 100 simulated data were generated. The 100 replications are commonly used in DIF simulation studies, and it has been shown that the sampling variability can be controlled (e.g., Bolt, 2002; Stark et al., 2006; Wang et al., 2012; Woods & Grimm, 2011). We set the total test length at 20, given that 20 item tests are commonly used in both educational and psychological settings (We also included the 40-item test conditions for Type I error only in accordance with the suggestion of an anonymous reviewer). In addition, we chose a relatively wider range of sample sizes and DIF sizes because the focus of this study is to investigate the performance (i.e., Type I error and power) of newly proposed DIF methods. We expect that the DIF detection power would increase as sample size and DIF size increased.

Data Generation

The simulated data were separately generated for the reference and focal groups. Depending on the subpopulation trait distributions, the examinees' latent abilities were randomly generated from a normal distribution with the same or different subpopulation mean. For the equivalent group condition, both groups were drawn from $N(0, 1)$, and for the nonequivalent group condition, the reference group was drawn from $N(0, 1)$ and the focal group was drawn from $N(1, 1)$. We included the different subpopulation mean conditions to assess the performance of the DIF detection methods when the means of the latent trait distribution are different across the reference and focal groups (i.e., impact). Previous studies have shown that the impact condition could inflate the Type I error of DIF detection methods (Finch, 2005; Wang et al., 2013). To generate simulated data, true discrimination (α) and difficulty (β) item parameters of the reference group were randomly generated from $U(.5, 2)$ and $U(-2, 2)$, respectively. For non-DIF items, the same item parameters were used for the focal group. For DIF items, item parameters of the focal group were manipulated based on DIF generating conditions. For example, for the condition where 10% nonuniform DIF with .2 DIF size, the discrimination parameters of randomly selected DIF items (2 out of 20) were shifted by adding .2 to the reference item parameters ($\alpha_{DIF}^F = \alpha_{DIF}^R + .2$). Similarly, for the condition where the 20% uniform DIF with 1.0 DIF size, the difficulty parameters of randomly selected DIF items (4 out of 20) were shifted by adding 1.0 to the reference item parameters ($\beta_{DIF}^F = \beta_{DIF}^R + 1.0$).

In this study, we considered the discrimination parameter differences of .2, .4, and .6 as small, medium, and large for the nonuniform DIF conditions and the difficulty parameter differences of .2, .6, and 1.0 as small, medium, and large for the uniform DIF conditions. These item parameter differences have been adapted in previous DIF studies including two group settings (e.g., Kim & Cohen, 1992; Oshima et al., 1997) and many group settings (e.g., Rutkowski & Svetina, 2014). For example, in Programme for International Student Assessment (PISA) 2018 reading assessment, the item parameter differences for the discrimination and difficulty parameters across the participating countries ranged from .01 to .88 and from .05 to 1.17, respectively (Joo et al., 2021; OECD, 2019). In addition, we also considered the 10% and 20% DIF conditions because the proportions are commonly found in practice (e.g., Joo et al., 2021; Lee et al., 2021; Stark et al., 2004, 2006) and research settings (e.g., Kim & Cohen, 1992, 1998; Oshima et al., 1997, 2006; Rutkowski & Svetina, 2014).

It is worthwhile to note that we randomly generated item parameters of the reference group for each replication to reduce the impact of item parameters on the DIF detection. Note that the item characteristic curve (ICC) difference could be different depending on the item parameters. For example, the ICC difference in the center of the latent trait continuum is more evident than those in the extreme ends. Once the item and ability parameters were generated for each group, response data were separately generated first, then combined into a whole dataset for MCMC multigroup estimation.

Model Estimation

Item and ability parameters were estimated using MCMC multigroup estimation. More specifically, the Metropolis-Hasting (MH) within Gibbs sampling (Patz & Junker, 1999) was developed. For the discrimination and difficulty parameters, we specified $LN(0, .25)$ and $N(0, 1)$ as the prior distributions for all items. Note that the prior distributions for the discrimination and difficulty parameters are commonly used in MCMC estimation (e.g., Béguin & Glas, 2001; Fox & Glas, 2001). Similarly, for the ability parameters, we used $N(0, 1)$ as the prior distribution for all examinees. For the focal group mean and variance parameters (μ_{θ_F} and $\sigma_{\theta_F}^2$), we used the noninformative hyperprior distributions. Previous studies have discussed that group population parameters are more challenging to estimate and a misspecified prior could result in biased parameter estimates (Kleijn & van der Vaart, 2006; Gelman et al., 2013; Gifford & Swaminathan, 1990). In addition, Gelman (2006) compared several priors including inverse-gamma, Cauchy, and uniform distributions for a variance parameter and found that the uniform distribution showed acceptable performance, in the sense that it has minimal impact on posterior inferences. Based on these findings, we chose the uniform distribution as a hyperprior for the population distribution parameters and consequently, the probability was dropped out from the MH within Gibbs sampling algorithm (see the Appendix 1 for a more detailed explanation). For the scale specification, we fixed the mean and variance of the reference group to be 0 and 1.

Prior to the MCMC estimation, initial values should be specified for all parameters. The initial values for the discrimination and difficulty parameters were specified as 1 and 0, respectively, for all items. The initial values for the ability parameters were specified as 0 for all examinees. Similarly, for the focal group mean and variance parameters, 0 and 1 were specified.

For the MH within Gibbs algorithm, we set the total number of iterations at 5,000 and only the last 3,000 samples were saved to construct the posterior distribution (i.e., burn-in period = 2000). We also replicated the iterations three times for each parameter (i.e., number of chains = 3), so the total number of posterior samples was 9,000. The final model parameter estimates were obtained by averaging the posterior samples. The total number of iterations, burn-in period, and chains was determined based on convergence criterion $\hat{R}$ statistic (Gelman & Rubin, 1992). We used the $\hat{R}$ statistics criterion of 1.2 to ensure the convergence of the MCMC algorithm (Gelman et al., 2013).

DIF Detection

To detect DIF using the PPMC approach, item parameters were estimated with an equal constraint across groups. The item parameter estimates were then evaluated with the PPMC method using the discrepancy statistics (infit, outfit, OSD, and Q1). Specifically, we computed the PPP values for each item by group and flagged an item as DIF if the PPP value from one of the groups was greater than an upper limit or less than a lower limit. As discussed in the introduction, because the PPP distribution generally does not follow a uniform distribution under the null condition, PPMC hypothesis testing with the nominal α level produces the Type I error rates less than

α (Levy et al., 2009; Sinharay et al., 2006). To examine the behavior of PPP values and obtain comparable results to the existing methods in the context of DIF analysis, the empirical PPP distributions for all discrepancy statistics across sample size were examined under the null condition (i.e., no DIF condition).

In the simulation, we also included Lord's chi-square (Lord, 1980) and Raju's unsigned area (Raju, 1990) DIF methods in simulation. Before conducting Lord and Raju DIF analysis, item parameters were separately estimated with marginal maximum likelihood (MML) estimation. Two separate item parameters from the reference and focal groups were then linked via mean-mean anchoring (Cook & Eignor, 1991). Once the MML item parameter estimates ($\hat{\alpha}_j^R$, $\hat{\alpha}_j^F$, $\hat{\beta}_j^R$, and $\hat{\beta}_j^F$) were obtained on the same metric, Lord's chi-square was computed as:

$$\text{Lord } \chi_j^2 = \mathbf{v}_j' \mathbf{\Gamma}_j^{-1} \mathbf{v}_j, \quad (10)$$

where $\mathbf{v}_j = [\hat{\alpha}_j^R - \hat{\alpha}_j^F, \hat{\beta}_j^R - \hat{\beta}_j^F]'$. $\mathbf{\Gamma}_j^{-1}$ is the inverse of the asymptotic sampling variance-covariance matrix. Under the null hypothesis, Lord's chi-square statistic asymptotically follows the chi-square distribution with two degrees of freedom. In addition, Raju's exact unsigned area (EUA) was computed as:

$$\text{Raju EUA} = H_j = \left| \frac{2(\hat{\alpha}_j^F - \hat{\alpha}_j^R)}{\hat{\alpha}_j^R \hat{\alpha}_j^F} \ln \left(1 + \exp \left[\frac{\hat{\alpha}_j^R \hat{\alpha}_j^F (\hat{\beta}_j^F - \hat{\beta}_j^R)}{\hat{\alpha}_j^F - \hat{\alpha}_j^R} \right] \right) - (\hat{\beta}_j^F - \hat{\beta}_j^R) \right|. \quad (11)$$

Raju (1990) showed that the H_j statistic can be used to detect DIF with the test statistic $Z = H_j / \text{Var}(H_j)^{1/2}$, which asymptotically follows the standard normal distribution. Previous studies, however, reported that both Lord and Raju DIF methods inflate Type I error when sample size increases, and the Lord DIF method showed better performance than the Raju DIF method in terms of detecting correct DIF (power; Cohen & Kim, 1993; McLaughlin & Drasgow, 1987). In this study, the Lord and Raju DIF methods served as the "reference methods" to the PPMC DIF methods. Because PPMC DIF methods were newly proposed and their performance was unknown under various conditions, it is important to establish reference criteria with the existing methods.

Data generation, model estimation, and PPMC DIF computation were simultaneously performed using the computer software language Ox (Doornik, 2009). In addition, Lord and Raju DIF analyses were performed separately with the "difR" package (Magis et al., 2012) implemented in the statistical software R (R Core Team, 2020). In addition, the Ox and R codes are publicly available in the GitHub repository (<https://github.com/seunghwan-joo/PPMC-DIF.git>).

Outcome Variables

To evaluate the performance of the PPMC DIF methods, we computed Type I error and power to detect DIF. Type I error was computed as the number of replications in which non-DIF items were flagged as DIF (i.e., false positive). Similarly, power to detect DIF was computed as the number of replications in which DIF items were flagged as DIF (i.e., hit rate). The proportions across replications were computed for each item first and then averaged across non-DIF items (Type I error) and DIF items (power), respectively, to summarize the results.

Results

PPP Distribution

Figure 1 shows the PPP distributions for the discrepancy statistics across sample sizes. The PPP values were computed under the null condition; all item responses were generated from the 2PL model and no DIF was included. For each discrepancy statistic and sample size, the distributions were created from the PPP values obtained from 20 items with 100 replicated data (i.e., a total of 2,000 PPP values). As shown in Figure 1, the PPP distributions across the sample sizes were almost identical, indicating that the different sample sizes did not affect the PPP distributions. More importantly, the PPP values from infit and outfit ranged from 0 to 1 and formed a symmetric distribution around .5, whereas the PPP values from OSD and Q1 formed a left-skewed distribution with a mode of .5.

This finding first confirms that the PPP values do not follow the uniform distribution and consequently, the hypothesis test with PPP would produce the Type I error less than α . Second, this result indicates that the hypothesis testing with PPP should be conducted differently across discrepancy statistics. In the context of PPMC DIF detection, we conducted a two-tailed hypothesis test for infit and outfit (i.e., an item is flagged as DIF if the PPP value is greater than .975 or less than .025) and a one-tailed hypothesis test for OSD and Q1 (i.e., an item is flagged as DIF if the PPP value is less than .05). These criteria were similar to a classical hypothesis test with $\alpha = .05$; however, we expect Type I error would be smaller than .05. Note that the decision for two types of hypothesis testing was also based on previous PPMC studies (e.g., Levy et al., 2009). For Lord and Raju statistics, we used .05 as the nominal level for DIF detection.

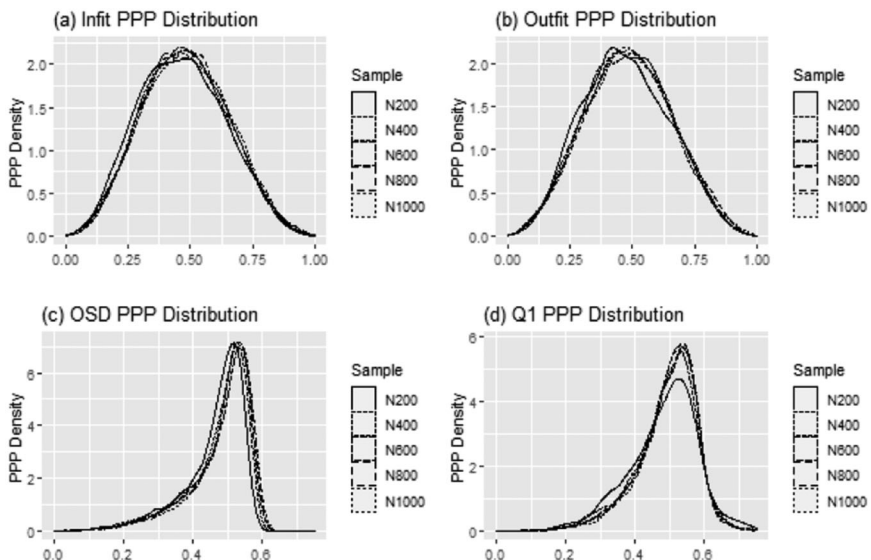


Figure 1. The PPP distributions of the discrepancy statistics across sample sizes.

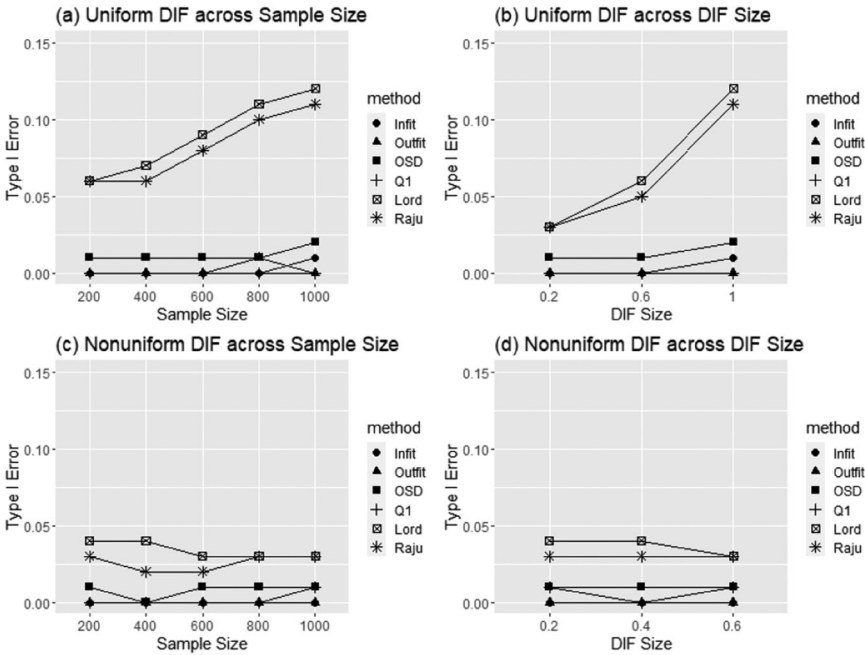


Figure 2. Type I error of the DIF methods for detecting uniform and nonuniform DIF across sample size and DIF size.

Type I Error

Figure 2 illustrates the Type I error results for the uniform DIF conditions (panels a and b) and the nonuniform DIF conditions (panels c and d). The Type I error in Figure 2 was created under the condition where DIF percentage was 10% and equivalent subpopulation distributions. In addition, DIF sizes were fixed at 1 and .6 (large DIF) for the uniform and nonuniform DIF conditions, respectively, for Figure 2 panels a and c. Similarly, the sample size was fixed at 1,000 for the uniform and nonuniform DIF conditions for Figure 2, panels b and d.

Overall, the PPMC DIF methods controlled Type I error reasonably better for both uniform and nonuniform DIF than the Lord and Raju methods. As shown in Figure 2, the Type I error of the PPMC DIF methods was lower than .05 across all conditions. The Lord and Raju DIF methods also showed well-controlled Type I error results in the nonuniform DIF conditions. More specifically, for the nonuniform DIF condition, Type I errors of PPMC, Lord, and Raju methods were lower than the nominal level (.05) across sample size and DIF sizes. However, for the uniform DIF condition, the Lord and Raju DIF methods considerably inflated Type I error as sample size and DIF size increased. For example, Type I errors were .12 and .11 for the Lord and Raju DIF methods when the sample size was 1,000 and the DIF size was 1.0. In contrast, the PPMC DIF methods showed better-controlled Type I error for the uniform DIF condition. Type I errors were below .025 across sample sizes and DIF sizes.

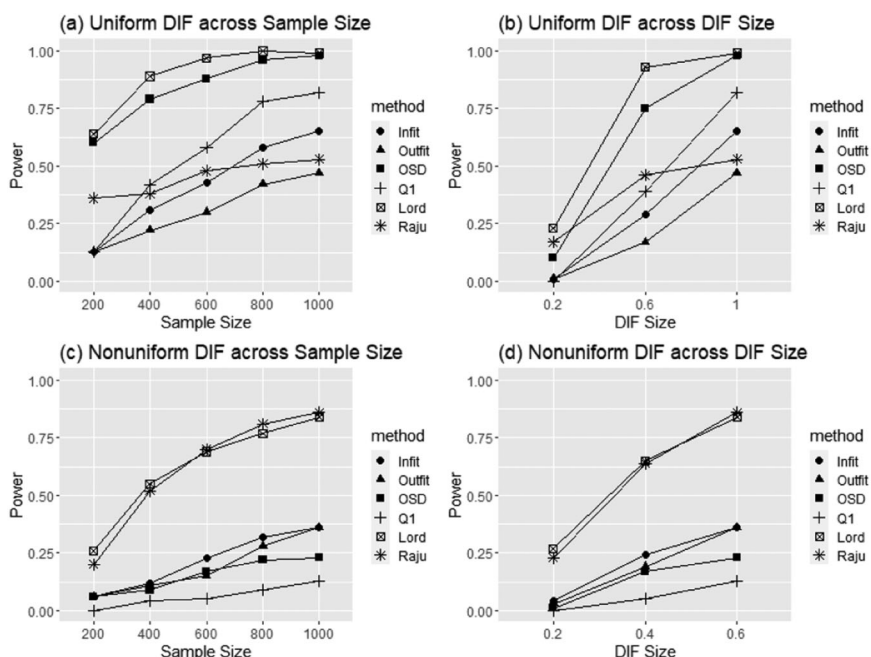


Figure 3. Power of the DIF methods for detecting uniform and nonuniform DIF across sample size and DIF size.

Power

Figure 3 illustrates the power results for the uniform DIF conditions (panels a and b) and the nonuniform DIF conditions (panels c and d). Figure 3 was created under the same conditions as Figure 2. Overall, power to detect DIF increased substantially as sample size and DIF size increased for all methods. For the uniform DIF conditions, PPMC with the OSD statistic and the Lord method showed similar results and outperformed the other DIF methods. Power for PPMC with OSD and Lord methods was close to 1 when the sample size was 1,000 and the DIF size was 1. PPMC with infit and outfit statistics did not show acceptable performance (power < .80) across the conditions, and surprisingly, the Raju method showed low performance for the uniform DIF condition. For the nonuniform DIF condition, Lord and Raju methods overall outperformed the PPMC DIF methods. The power for Lord and Raju methods was close to .80 when the sample size was 1,000 and the DIF size was 1, whereas the corresponding values for the PPMC methods did not reach the acceptable criterion (power > .80). Infit and outfit statistics showed slightly better performance than OSD and Q1 statistics for the nonuniform DIF condition.

DIF Percentage and Subpopulation Distribution

We also varied the DIF percentages from 10% to 20% and included two types of subpopulation distributions (equivalent and nonequivalent). Figure 4 represents Type I error and power results for different DIF percentage conditions across sample sizes,

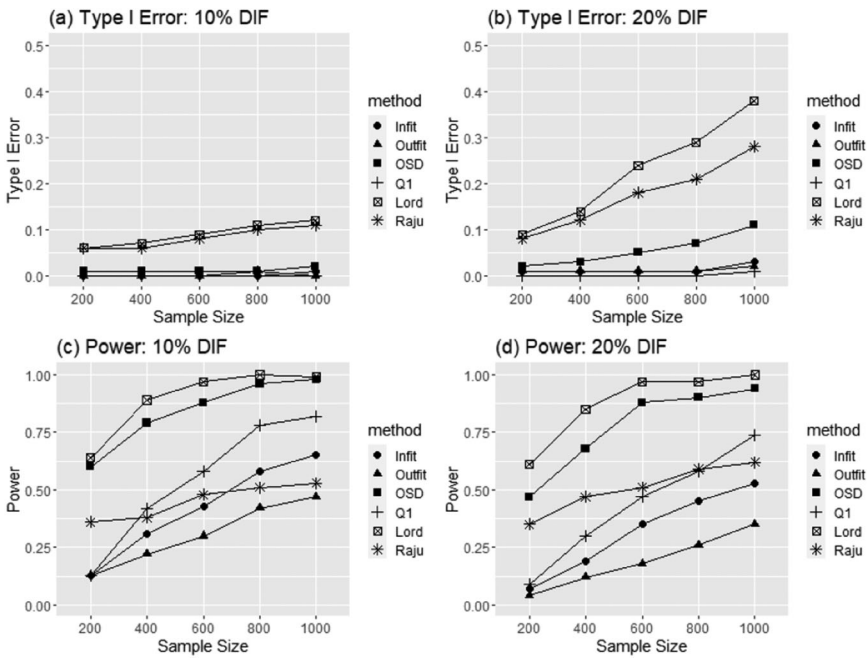


Figure 4. Type I error and power to detect uniform DIF for 10 and 20 percentages DIF across sample size.

and Figure 5 presents Type I error and power results for different subpopulation distribution conditions across sample sizes. Both Figures 4 and 5 results were obtained when DIF size was 1.0 (large DIF) and uniform DIF was considered.

As shown in Figure 4, increasing DIF percentage notably increased Type I error for the Lord, Raju, and PPMC DIF methods. Consistent with the previous Type I error results, Lord and Raju showed the most substantial Type I error inflation as sample size increased, whereas the PPMC DIF methods showed reasonable Type I error across sample size. More specifically, PPMC with the OSD statistic showed inflated Type I error (i.e., greater than the nominal level) when the sample size was 1,000 and the DIF size was 1.0. Note that the Type I error inflation for the Lord and Raju methods is expected due to the increment of the anchor item contamination. Previous studies also reported that Type I error is frequently inflated when the proportion of DIF increases (Stark et al., 2006; Wang et al., 2012; Wang & Yeh, 2003; Woods, 2009b). In terms of power, Lord and PPMC with the OSD statistics showed similar comparable results across two DIF percentage conditions. Considering the Type I error inflation of the Lord and Raju methods, PPMC with the OSD statistic showed more stable performance across the 10% and 20% DIF conditions. Similar to the previous power results of the Raju and PPMC with the infit, outfit, and Q1 statistics showed less power to detect uniform DIF across two DIF percentage conditions.

In addition, Figure 5 represents the Type I error and power results across different subpopulation distributions for the uniform DIF condition. Consistent with the previous results, Lord and Raju increased Type I error as sample size increased, and Type

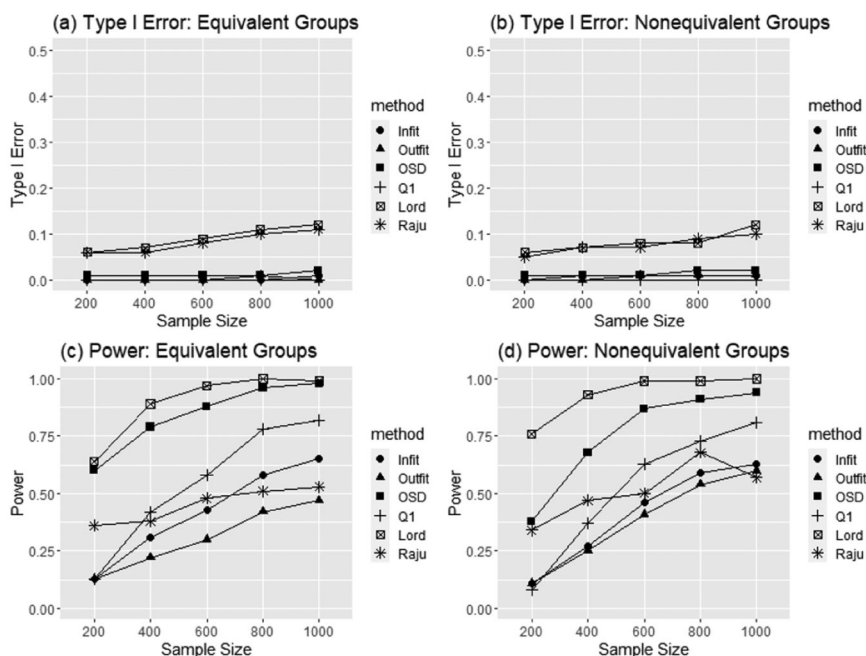


Figure 5. Type I error and power to detect uniform DIF for the subpopulation distributions across sample size.

I error from the PPMC DIF methods remained consistent as sample size increased in both equivalent and nonequivalent group conditions. Furthermore, the power of all DIF methods also showed similar results across different subpopulation distribution conditions.

Type I Error for the 40-Item Test

Given that we observed the inflated Type I error for the Lord and Raju DIF methods, we investigated an additional simulation condition where the number of items was extended to 40. Previously, Uttaro and Millsap (1994) examined the performance of the MH method for both short (20 items) and longer (40 items) test lengths. For the 20-item test, the MH procedure gave an inflated Type I error rate when the groups differed in ability distributions. However, the inflation in the Type I error rate diminished entirely in the 40-item test. Although we considered 20 items in the main simulation design because it is commonly used, it does not adequately capture longer high-stakes tests such as those found in certification and licensure. Thus, we additionally investigated the Type I error of the DIF methods under the 40 item-test conditions. We fixed the condition where the subpopulation distributions were different across two groups (impact) and we focused on uniform DIF.

Figure 6 shows the Type I error rates of PPMC DIF and existing parametric DIF methods (Lord and Raju) for the 40-item test. The first row (panels a and b) represented Type I error for the 10% DIF condition (i.e., 4 items were DIF out of 40 items)

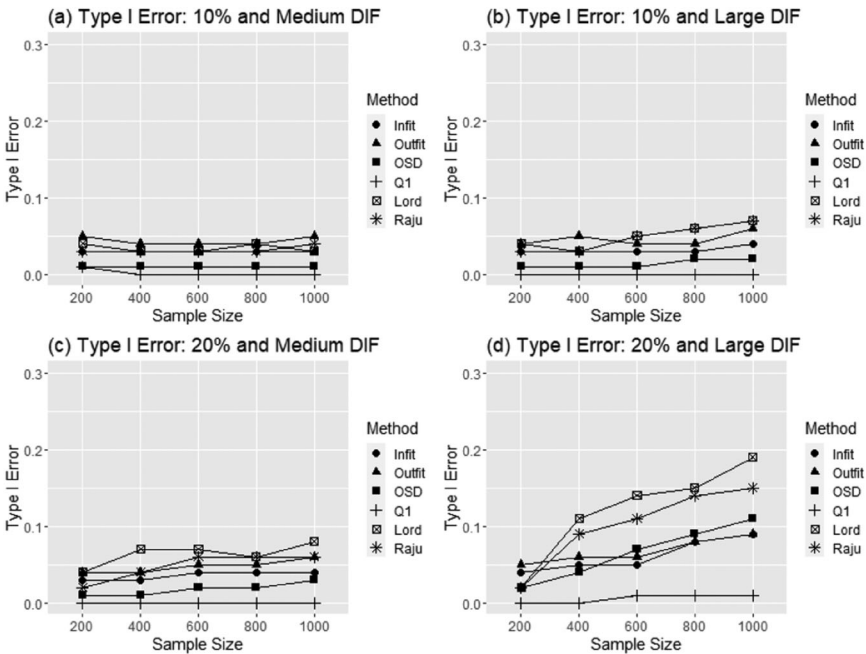


Figure 6. Type I error of the DIF detection methods for the 40-item test across sample size.

and the second row (panels c and d) represents Type I error for the 20% DIF condition (i.e., 8 items were DIF out of 40 items). In addition, the left column indicates results for the medium size DIF (i.e., .6 difference for the difficulty parameters) and the right column indicates results for the large size DIF (i.e., 1.0 difference for the difficulty parameters).

Overall, we found that Type I errors were better controlled for the 40-item test than the 20-item test. When 10% of the items were generated with medium DIF, PPMC, Lord, and Raju methods controlled Type I error under the nominal level (.05) as sample size increased. However, as the size and the percent of DIF increased, the Lord and Raju DIF methods showed inflated Type I error results. Although the inflation was not as substantial as the 20-item test, the Type I error of Lord and Raju methods was higher than the nominal level when the large and 20% of DIF were considered. For example, for the same condition, Type I errors were .19 and .15 for the Lord and Raju methods whereas PPMC DIF with the infit, outfit, OSD, and Q1 statistics were .09, .09, .11, and .01, respectively.

Real Data Example

To evaluate the efficacy of the PPMC DIF methods in applied settings, we analyzed real participant data. We used an open-source personality data set (openpsy-chometric.org) collected from the Big Five personality measure. For this demonstration, we chose an extraversion scale that consists of 20 items. The extraversion test

was originally constructed as five-point scales, but we dichotomized the responses in which any response equal to 4 or 5 was recoded as 1, and 0 otherwise. Note that we dichotomized the data only for the demonstration purpose but readers should be cautious about the dichotomization when two groups are compared (Rutkowski et al., 2019). We then randomly sampled 1,000 examinees from the male group and sampled another 1,000 examinees from the female group. Before conducting the DIF test across two groups, we first investigated the overall model fit for the unidimensional 2PL model. We fitted the example data with the unidimensional 2PL model and computed the commonly used absolute model fit statistics including the root mean square error of approximation (RMSEA), comparative fit index (CFI), Tucker-Lewis index (TLI), and standardized root mean squared residual (SRMSR). For the male group, RMSEA was .07, CFI was .97, TLI was .97, and SRMSR was .05. Similarly, for the female group, RMSEA was .07, CFI was .97, TLI was .96, and SRMSR was .05. The results were within the acceptable criteria (e.g., Hu & Bentler, 1999) indicating that the 2PL model fits the example data.

To apply PPMC DIF analysis, we estimated the data with 2PL using multigroup MCMC estimation. We constrained the 2PL item parameters to be equal across the groups and fixed the mean and standard deviation (SD) of the reference (male) group to be zero and one, respectively, to set the scale. The mean and SD of the focal (female) group were estimated. We also checked the convergence of the MCMC algorithm using $\hat{R}$ statistics. Once the parameters were estimated, we computed the PPP values for each item by group. In addition, using the same data set, we conducted Lord and Raju DIF tests to compare the performance of various DIF tests. Similar to the simulation study, we applied a two-tailed test for PPMC with infit and outfit and a one-tailed test for PPMC with OSD and Q1, Lord, and Raju methods.

Table 1 shows the item parameter estimates, their PSDs, and convergence statistics across items from the 2PL multigroup MCMC estimation. In addition, to provide more explicit information about convergence of MCMC estimation, we also provided autocorrelation plots and Monte Carlo Standard Errors (MCSE) in the Appendix 2 (Figure B1 and Table A2.1). Overall, the item parameters from 2PL showed reasonable values, ranging from .79 to 2.20 for the a -parameters and from -2.85 to 2.58 for the b -parameters. The PSD values for the parameter estimates were also acceptable, ranging from .10 to .24 for the a -parameters and from .04 to .09 for the b -parameters. The convergence criterion, $\hat{R}$ statistics confirmed the practical convergence of the MCMC algorithm ($\hat{R} < 1.2$; Gelman et al., 2013). Note that the item parameters from the MCMC estimation were comparable to the MML estimation indicating that item parameter estimates from both estimation algorithms were comparable. We reported the item parameter comparison in the Appendix 3 (Table A3.1).

Table 2 shows the p values from the various DIF methods. For the PPMC DIF method, only item 16 was detected as DIF using infit and outfit, items 2, 9, 12, 16, 18, and 20 were detected as DIF using OSD, and items 16, 18, and 20 were detected as DIF using Q1 as the discrepancy statistic. In contrast, for the conventional parametric DIF approaches, items 2, 9, 11, 12, 16, 18, and 20 were detected as DIF using the Lord's chi-square method and items 2, 3, 9, 11, 12, 13, 16, 18, 19, and 20 as DIF using the Raju's area method. Note that PPMC with infit and outfit only detected one DIF item. This finding highlights the simulation study result where infit and outfit

Table 1
Item Parameter Estimates, Posterior Standard Deviation, and Convergence Statistics from the Real Data Example

Item	Estimates		PSD		$\hat{R}$ Statistics	
	<i>a</i> -param	<i>b</i> -param	<i>a</i> -param	<i>b</i> -param	<i>a</i> -param	<i>b</i> -param
1	1.75	.30	.14	.05	1.00	1.03
2	1.18	−2.00	.13	.05	1.01	1.01
3	1.33	−.04	.12	.04	1.01	1.01
4	1.22	−1.45	.15	.04	1.02	1.01
5	.88	−.37	.17	.04	1.00	1.02
6	2.20	−1.53	.13	.06	1.01	1.00
7	1.64	−1.77	.19	.04	1.01	1.05
8	1.41	1.67	.10	.05	1.00	1.01
9	1.45	−.81	.10	.05	1.03	1.02
10	.99	.75	.15	.04	1.01	1.01
11	1.23	−2.38	.24	.04	1.00	1.03
12	1.34	−.99	.13	.05	1.00	1.03
13	1.23	.00	.21	.04	1.01	1.04
14	.86	1.04	.19	.08	1.00	1.00
15	1.06	2.58	.13	.05	1.01	1.02
16	.79	.47	.13	.09	1.02	1.00
17	.89	1.97	.14	.05	1.00	1.05
18	1.33	−.33	.13	.07	1.01	1.00
19	1.31	−2.86	.19	.04	1.00	1.03
20	1.16	1.77	.14	.07	1.00	1.00

Note. *a*-param = item discrimination parameter, *b*-param = item difficulty (location) parameter.

showed low power to detect DIF. We also found that the PPMC with OSD and Lord methods showed similar results except for item 11. This finding is consistent with the simulation study results where the PPMC with OSD and Lord DIF methods have similar power to detect DIF. Finally, the Raju methods detected most of the items as DIF. However, we believe that the results might also be inflated based on the simulation study results.

Discussion

In this study, we proposed a Bayesian approach for detecting DIF under the PPMC framework and compared the performance of various discrepancy statistics under simulation conditions. We considered four item fit statistics (infit, outfit, OSD, and Q1) as discrepancy statistics for the PPMC DIF methods, and Type I error and power were evaluated with a wide range of sample sizes and DIF sizes. We also considered both uniform and nonuniform DIF, two DIF percentages, and equivalent and nonequivalent subpopulation distributions in the simulation design. Based on the simulation results, we found that the PPMC DIF methods produced well-controlled Type I error across the simulation conditions and, importantly, PPMC with the OSD

Table 2
p-Values of the PPCM, Lord, and Raju DIF Detection Results from the Real Data Example

PPMC DIF										
Item	Reference Group				Focal Group				Lord	Raju
	Infit	Outfit	OSD	Q1	Infit	Outfit	OSD	Q1		
1	.66	.36	.53	.46	.36	.59	.53	.50	.20	.08
2	.25	.22	.01	.07	.82	.62	.01	.07	.00	.00
3	.22	.40	.58	.45	.77	.63	.57	.47	.06	.01
4	.23	.19	.50	.25	.81	.74	.48	.49	.80	.52
5	.29	.32	.52	.44	.72	.68	.53	.54	.06	.02
6	.05	.16	.15	.24	.96	.89	.13	.32	.10	.21
7	.52	.41	.39	.47	.54	.31	.40	.36	.84	.56
8	.41	.46	.26	.46	.53	.60	.29	.41	.13	.06
9	.33	.53	.00	.04	.57	.71	.01	.05	.00	.00
10	.21	.37	.44	.44	.78	.68	.45	.52	.09	.08
11	.08	.22	.29	.18	.94	.88	.32	.28	.00	.00
12	.13	.20	.02	.10	.88	.87	.02	.19	.00	.00
13	.28	.32	.42	.41	.70	.72	.41	.39	.08	.02
14	.85	.77	.35	.43	.19	.15	.34	.35	.34	.48
15	.44	.35	.59	.51	.61	.58	.58	.54	.64	.37
16	.01	.00	.00	.00	.00	.00	.00	.00	.01	.00
17	.14	.18	.59	.37	.91	.83	.58	.36	.12	.05
18	.11	.06	.01	.00	.11	.06	.02	.00	.03	.01
19	.15	.19	.38	.30	.88	.84	.42	.37	.07	.01
20	.15	.09	.04	.00	.15	.09	.00	.00	.00	.00

Note. Values in bold indicate the *p* value less than .025 or greater than .975 for infit and outfit, and the *p* values less than .05 for OSD and Q1, Lord, and Raju DIF.

statistic detects uniform DIF with high power. Finally, we demonstrated the feasibility of the proposed DIF methods using the real participant data publicly available for researchers.

There are several implications of this study. First, PPCM could be effectively implemented for detecting uniform DIF under the Bayesian framework. Recently, the Bayesian estimation technique MCMC has been widely used in educational measurement research in general because of its flexibility and accuracy. If researchers are interested in scaling and scoring with MCMC for multiple group analysis, the PPCM DIF methods can be readily adapted using the posterior samples from the MCMC algorithm. Our study provided the procedure for the PPCM DIF methods and verified the performance of several discrepancy statistics. Based on the simulation study results, we recommend OSD as the discrepancy statistic for detecting uniform DIF.

In addition, we also compared the PPCM DIF methods with the existing DIF methods implemented in MML estimation. In the current study, we included Lord's chi-square and Raju's unsigned area for detecting DIF. In terms of Type I error results,

we found that the PPMC methods outperformed the existing methods. The PPMC DIF methods with OSD and Q1 statistics showed a better Type I error control than the Lord and Raju methods when uniform DIF was generated. As sample size and DIF size increased, the Lord and Raju methods showed the inflated Type I error results, whereas the PPMC with OSD and Q1 consistently controlled the Type I error close to the nominal level (.05). For the power results, the PPMC with OSD showed comparable performance as the Lord DIF method. In sum, the PPMC DIF methods can be alternatives to the existing DIF methods and could be used for large-scale assessments where a large sample size is often collected. However, it is worthwhile to note that the performances of the PPMC and existing DIF methods could be affected by the estimation algorithm (MML vs. MCMC). Although we confirmed that both estimation algorithms produced minimal bias for item parameter estimates and both estimates were comparable, we note that different estimation methods could still affect the DIF detection results.

In this study, we also investigated the PPP distributions for each discrepancy statistic. We found the PPP distribution showed similar shapes across different sample sizes. This finding is consistent with previous studies (Levy et al., 2009; Sinharay, 2005; Sinharay et al., 2006) and confirmed that PPP-values for detecting DIF are also conservative regardless of sample size. Note that theoretical and empirical studies have reported that the PPP-values generally have reasonable frequentist properties (Gelman et al., 1996; Sinharay, 2005) and the PPP-values often failed to reject the null hypothesis, even asymptotically for some choices of test statistics (Bayarri & Berger, 2000). In the current study, we found the PPP values from infit and outfit statistics shaped the symmetric distribution centered around .5 and ranged from 0 to 1. On the other hand, the PPP values from OSD and Q1 showed the left-skewed distribution and ranged from 0 to .75. Based on the PPP distributions, we considered one-tailed hypothesis testing with .05 as the α level for the OSD and Q1 statistics and two-tailed hypothesis testing for the infit and outfit statistics. Using these approaches, we found that the Bayesian hypothesis testing using PPP-values is comparable to the frequentist's hypothesis testing for detecting DIF. Type I error was controlled close to the nominal level and power was comparable to the existing methods. Based on this evidence, applied researchers and practitioners can conduct hypothesis testing using the PPP-values for the PPMC DIF detection. More importantly, we recommend researchers investigate the PPP distributions empirically if different discrepancy statistics were considered.

Finally, from the simulation results, we found that the PPMC DIF methods did not show satisfying performance for detecting nonuniform DIF across simulation conditions. Although Type I error was better controlled in the nonuniform DIF condition, power did not reach the acceptable criterion ($>.80$) across sample size and DIF size. Note the relatively low power for detecting nonuniform DIF could be problematic because the nonuniform DIF also occurs in practical settings although the size of nonuniform DIF is generally smaller than uniform DIF (e.g., Kim & Cohen, 1992; Oshima et al., 1997; Raju et al., 1995; Rutkowski & Svetina, 2014). Based on the finding, we recommend applied researchers carefully interpret DIF results. It is possible that the PPMC DIF methods with different discrepancy statistics could be more

effective for detecting nonuniform DIF, and thus future study is needed to explore alternative item fit statistics for nonuniform DIF.

It is worthwhile to note that the PPMC DIF methods, like the other parametric DIF methods, assume that observed responses fit the IRT model. This is a strong assumption for detecting DIF and the other applications because the PPP values were originally developed as a model data fit index. If the model does not fit the observed data, it is difficult to conclude that the extreme PPP value indicates DIF. It is true that in the context of multiple group analysis, DIF or measurement noninvariance is considered as a part of model misfit for the item equality constraint. Large-scale assessments such as PISA, for example, evaluate DIF by simultaneously imposing the equality constraint across the participating groups (Rutkowski et al., 2014). This approach is advantageous in the context of large-scale assessments because there are more than two participating groups, and the target populations are generally heterogeneous. The traditional DIF methods could be more burdensome and challenging for evaluating DIF with many groups (Rutkowski & Svetina, 2014). In this study, our focus was to detect DIF by evaluating equally constrained items across multiple groups. However, again, this approach assumes that the observed data adequately fit individual groups. Thus, applied researchers should evaluate model-data fit prior to the DIF detection using PPMC. Once the model data fit is confirmed for each group, the PPMC DIF analysis should be further conducted to ensure the results.

This study also includes several limitations. First, we only considered dichotomous responses in the current study, and polytomous responses were not explored. We believe the PPMC DIF methods can be extended for polytomous IRT models such as the generalized partial credit model (GPCM, Muraki, 1992) or nominal response model (NRM, Bock, 1972). Second, for detecting DIF, the constrained baseline approach was mainly investigated. In other words, all item parameters including studied items were constrained equal across groups. However, previous research reported that the free baseline approach is more effective for controlling Type I error, because the nonstudied items are freely estimated across groups so that the possibility of anchor item contamination can be substantially reduced (Stark et al., 2006). The free baseline approach can be alternatively implemented for the PPMC DIF methods by freeing nonstudied items while studied items are constrained to be equal.

Nonetheless, this study provides valuable information about how to conduct DIF analysis with the Bayesian PPMC approach, which is being considered as an effective alternative to the IRT model fit assessment. We hope the PPMC DIF methods proposed in this study expand possibilities with Bayesian applications for psychometric research.

Acknowledgments

The authors would like to thank Dr. Sandip Sinharay and three anonymous reviewers for the insightful suggestions.

Appendix

The Description of Multigroup MCMC Estimation

The likelihood equation of the 2PL model for the multigroup case is:

$$P(X|\theta, \alpha, \beta) = \prod_{g=1}^G \prod_{j=1}^J \prod_{i=1}^N P(X_{ijg}|\theta_{ig}, \alpha_{jg}, \beta_{jg})^{X_{ijg}} [1 - P(X_{ijg}|\theta_{ig}, \alpha_{jg}, \beta_{jg})]^{(1-X_{ijg})}, \quad (A1)$$

where X is the $N \times J$ response matrix, θ is the $N \times 1$ ability parameter vector, α and β are the $J \times 1$ item parameter vectors, and μ and σ^2 are the $G \times 1$ group population parameter vectors. From the likelihood equation, the posterior distribution can be obtained as:

$$P(\theta, \alpha, \beta, \mu, \sigma^2|X) \propto P(X|\theta, \alpha, \beta)P(\alpha)P(\beta)P(\theta|\mu, \sigma^2)P(\mu)P(\sigma^2), \quad (A2)$$

where $P(\alpha)$ and $P(\beta)$ are the prior distributions for the parameters α and β , $P(\theta|\mu, \sigma^2)$ is the prior distribution for the parameter θ , $P(\mu)$ and $P(\sigma^2)$ are hyperprior distributions for the parameters μ and σ^2 respectively. To approximate the posterior distribution, we used the Metropolis-Hasting (MH) within Gibbs sampling (Patz & Junker, 1999).

To conduct MH within Gibbs sampling, initial values of model parameters should be first specified: θ_0 , α_0 , β_0 , μ_0 , and σ_0^2 . The MH within Gibbs sampling then updates throughout the iterative process. For iteration t ($t = 1, \dots, T$), sample parameters θ_t^* , α_t^* , β_t^* , μ_t^* , and σ_t^{2*} are randomly drawn from proposed distributions. The sampled parameters are accepted with the probabilities:

$$P(\theta_t^*, \theta_{t-1}) = \min\left(\frac{P(X|\alpha_{t-1}, \beta_{t-1}, \theta_t^*)P(\theta_t^*|\mu_{t-1}, \sigma_{t-1}^2)}{P(X|\alpha_{t-1}, \beta_{t-1}, \theta_{t-1})P(\theta_{t-1}|\mu_{t-1}, \sigma_{t-1}^2)}, 1\right), \quad (A3)$$

$$P(\alpha_t^*, \alpha_{t-1}) = \min\left(\frac{P(X|\alpha_t^*, \beta_{t-1}, \theta_t)P(\alpha_t^*)}{P(X|\alpha_{t-1}, \beta_{t-1}, \theta_t)P(\alpha_{t-1})}, 1\right), \quad (A4)$$

$$P(\beta_t^*, \beta_{t-1}) = \min\left(\frac{P(X|\alpha_t, \beta_t^*, \theta_t)P(\beta_t^*)}{P(X|\alpha_t, \beta_{t-1}, \theta_t)P(\beta_{t-1})}, 1\right), \quad (A5)$$

$$P(\mu_t^*, \mu_{t-1}) = \min\left(\frac{P(\theta_t|\mu_t^*, \sigma_{t-1}^2)P(\mu_t^*)}{P(\theta_t|\mu_{t-1}, \sigma_{t-1}^2)P(\mu_{t-1})}, 1\right), \quad (A6)$$

$$P(\sigma_t^{2*}, \sigma_{t-1}^2) = \min\left(\frac{P(\theta_t|\mu_t, \sigma_t^{2*})P(\sigma_t^{2*})}{P(\theta_t|\mu_t, \sigma_{t-1}^2)P(\sigma_{t-1}^2)}, 1\right). \quad (A7)$$

Note that θ_t , α_t , β_t , μ_t , and σ_t^2 are sequentially updated at iteration t using Equations A3–A7. The algorithm continues until the number of iterations are reached the predetermined maximum iteration ($t = T$) and repeats across chains ($c = 1, \dots$,

C). The final parameter estimates and their posterior standard deviation (PSD) are summarized after discarding samples from the burn-in period ($t = 1, \dots, B$):

$$\hat{\theta} = \frac{1}{C(T-B)} \sum_c \sum_{t=B+1}^T \theta_{ct}, \quad (\text{A8})$$

$$PSD(\theta) = \sqrt{\frac{1}{C(T-B-1)} \sum_c \sum_{t=B+1}^T (\theta_{ct} - \hat{\theta})^2}, \quad (\text{A9})$$

where θ_{ct} represents the model parameters for chain c at iteration t from the MH within Gibbs algorithm.

Appendix 2

Convergence of Multigroup MCMC Estimation

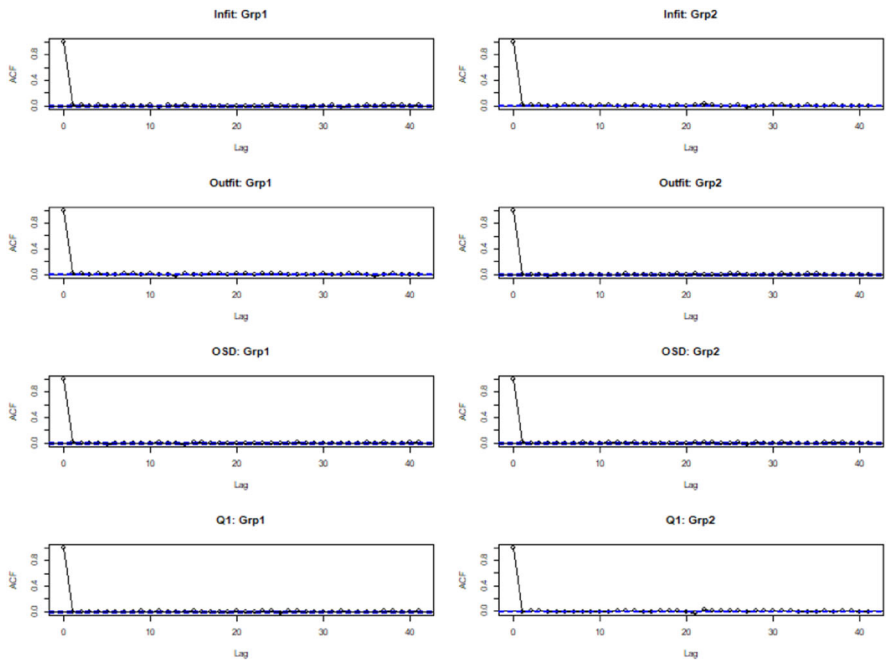


Figure B1. Autocorrelation plots of Markov chain samples for discrepancy statistics. [Color figure can be viewed at wileyonlinelibrary.com]

Note. that the blue lines indicates the 95% confidence intervals.

Table A2.1. *Monte Carlo Standard Errors (MCSE) for Discrepancy Statistics across Items and Groups*

Item	Infit		Outfit		OSD		Q1	
	Group 1	Group 2	Group 1	Group 2	Group 1	Group 2	Group 1	Group 2
1	.0004	.0004	.0002	.0001	.0006	.0006	.0004	.0004
2	.0004	.0004	.0001	.0001	.0002	.0002	.0004	.0004
3	.0003	.0003	.0001	.0001	.0004	.0003	.0004	.0004
4	.0004	.0003	.0002	.0001	.0003	.0003	.0004	.0004
5	.0004	.0004	.0002	.0002	.0003	.0003	.0004	.0004
6	.0004	.0004	.0001	.0001	.0002	.0002	.0004	.0004
7	.0004	.0004	.0003	.0003	.0004	.0004	.0004	.0004
8	.0003	.0003	.0001	.0001	.0004	.0004	.0004	.0004
9	.0003	.0003	.0001	.0001	.0005	.0005	.0004	.0004
10	.0004	.0003	.0001	.0001	.0004	.0004	.0004	.0004
11	.0003	.0003	.0003	.0002	.0004	.0003	.0004	.0004
12	.0003	.0003	.0004	.0004	.0003	.0003	.0004	.0004
13	.0003	.0003	.0002	.0002	.0003	.0003	.0004	.0004
14	.0004	.0004	.0003	.0003	.0002	.0002	.0004	.0004
15	.0004	.0004	.0001	.0001	.0004	.0003	.0004	.0004
16	.0002	.0002	.0006	.0009	.0003	.0003	.0004	.0004
17	.0002	.0002	.0006	.0005	.0003	.0003	.0004	.0004
18	.0004	.0004	.0005	.0004	.0002	.0002	.0004	.0004
19	.0003	.0003	.0003	.0003	.0002	.0002	.0004	.0004
20	.0004	.0004	.0001	.0001	.0005	.0005	.0004	.0004

Appendix 3

Table A3.1. *Item Parameter Estimates for Empirical Example Using MCMC and MML Estimation*

Item	MCMC		MML	
	<i>a</i> -param	<i>b</i> -param	<i>a</i> -param	<i>b</i> -param
1	1.75	.30	1.64	.31
2	1.18	−2.00	1.02	−2.28
3	1.33	−.04	1.41	−.11
4	1.22	−1.45	1.10	−1.57
5	.88	−.37	.91	−.43
6	2.20	−1.53	2.21	−1.59
7	1.64	−1.77	1.55	−1.79
8	1.41	1.67	1.29	1.83
9	1.45	−.81	1.47	−.83
10	.99	.75	1.02	.76

11	1.23	-2.38	1.10	-2.51
12	1.34	-.99	1.54	-.92
13	1.23	.00	1.23	.05
14	.86	1.04	.90	.90
15	1.06	2.58	1.15	2.38
16	.79	.47	.77	.41
17	.89	1.97	.88	2.03
18	1.33	-.33	1.26	-.41
19	1.31	-2.86	1.30	-2.83
20	1.16	1.77	1.13	1.76

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